

Exploratory Data Analysis of E-Commerce Sales Data

Executive Summary

This project analyzed 1,200 e-commerce transactions to identify sales trends, product performance, customer payment preferences, referral channel effectiveness, and revenue patterns. The analysis revealed that Chairs generated the highest revenue (\$195,620.11), Instagram was the most profitable referral source (\$275,285.45), and Online payments were the most frequently used payment method (258 transactions).

The findings provide actionable insights for inventory management, marketing optimization, and revenue growth strategies.

Dataset Overview

The dataset contains:

- 1,200 customer orders
- 14 business-related variables
- Transaction dates ranging from January 2023 to June 2025

Key Statistics

Metric	Value
Total Orders	1,200
Average Order Value	\$1,053.97
Median Order Value	\$823.62
Maximum Order Value	\$3,456.40
Minimum Order Value	\$11.39
Average Quantity Purchased	2.95 units

Product Revenue Analysis

Revenue generated by product category:

Product Revenue (\$)	
Chair	195,620.11
Printer	195,612.61
Laptop	192,126.56

Product Revenue (\$)

Tablet	186,568.95
Monitor	175,651.41
Desk	167,459.93
Phone	151,722.39

Insight

- Chairs generated the highest revenue at **\$195,620.11**.
- Phones generated the lowest revenue at **\$151,722.39**.
- The revenue gap between the highest and lowest performing products was approximately **\$43,897.72**.

Recommendation

Increase promotional focus on high-performing products such as Chairs, Printers, and Laptops while investigating pricing, customer demand, and marketing strategies for Phones.

Order Status Analysis

Status	Orders
Cancelled	250
Returned	247
Pending	237
Shipped	235
Delivered	231

Insight

A combined total of **497 orders** were either cancelled or returned, representing a significant operational concern that may affect customer satisfaction and profitability.

Recommendation

Conduct root-cause analysis on cancellations and returns to identify issues related to product quality, delivery performance, or customer expectations.

Payment Method Analysis

Payment Method Transactions

Online	258
Cash	246
Credit Card	234
Debit Card	232
Gift Card	230

Insight

Online payments accounted for the highest number of transactions (**258 orders**), indicating a strong customer preference for digital payment channels.

Recommendation

Continue investing in secure and frictionless online payment systems to improve customer experience and conversion rates.

Referral Source Performance

Referral Source Revenue (\$)

Instagram	275,285.45
Email	261,808.55
Google	250,441.48
Facebook	250,410.90
Referral	226,815.58

Insight

Instagram generated the highest revenue contribution at **\$275,285.45**, outperforming Referral traffic by **\$48,469.87**.

Recommendation

Increase marketing investment in Instagram campaigns while analyzing the factors contributing to its superior performance.

Revenue Trend Analysis

Monthly sales revenue fluctuated throughout the study period.

Examples from the analysis:

Month	Revenue (\$)
Jan 2023	56,685.75
Feb 2023	40,117.66
Mar 2023	48,609.37
Apr 2023	27,751.71
May 2023	63,836.84

Insight

Revenue variability suggests the influence of seasonal demand, promotions, or customer purchasing cycles.

Recommendation

Implement periodic sales forecasting models to improve inventory planning and marketing campaign timing.

Outlier Analysis

The TotalPrice distribution revealed several unusually high-value transactions, with the maximum transaction value reaching **\$3,456.40**.

Business Impact

These high-value purchases may represent premium customers or bulk orders and could form the basis of a targeted customer retention strategy.

Conclusion

The analysis identified clear revenue drivers and customer behavior patterns. Chairs generated the highest product revenue (\$195,620.11), Instagram was the most effective referral channel (\$275,285.45), and Online payments were the preferred transaction method (258 orders). These

insights can help management optimize product strategy, marketing investments, and operational decision-making to support sustainable business growth.